OVER A FLAT FINITELY PRESENTABLE TARGET MAP, RELATIVE PSEUDO-COHERENCE IS ORDINARY PSEUDO-COHERENCE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10859-10869 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 10871-10876, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Cross-chapter dependency satisfied: L16132 assembled last wave.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Etale-local definitions of absolute and relative pseudo-coherence on spaces; More on Morphisms, lemma check-relative-pseudo-coherence-on-charts (the schemes dictionary, card L16132).

Source reference. Original lemma 09UU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10876–10876. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let E in $D_{QCoh}(\mathcal{O}_X)$. If f is flat and locally of finite presentation, then the following are equivalent*

(1) *E is pseudo-coherent relative to Y, and*
(2) *E is pseudo-coherent on X.*

Proof. By étale localization and the definitions we may assume X and Y are schemes. For the case of schemes this follows from More on Morphisms, Lemma 09UU. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.